

CRF Spatial Correlation Field

$C(r) = \text{Cov}(N_i, N_j \mid d_{ij} = r)$: Structure, Sign, and the Gap

Short-range attraction, long-range repulsion, and the conservation law that writes the field

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Abstract

The CRF Fluctuation Law v3 established $Q_{\text{true}} = \sigma_N / \sqrt{N} \approx 1 / \sqrt{\kappa n T_{\text{avg}}}$ with a 5.2% gap. That gap requires an implied global covariance $\text{Cov}_{\text{implied}} \approx -0.025 \times \text{Var}$. Previous measurement found $\text{Cov}_{\text{all}} = -6.6 \times 10^{-4}$, 40 $\times$ too small, falsifying mean-field explanations.

This paper reports the spatial correlation field $C(r) = \text{Cov}(N_i, N_j \mid d_{ij} = r)$ measured at KL-graph distances $r = 1, \dots, 6$. The result: $C(1) = +0.0188$ (positive, nearest neighbours fire together), sign flip at $r = 3$, $C(5) = -0.0189$, $C(6) = -0.0703$ (strong long-range anti-correlation). The global weighted average $\text{Cov}_{\text{all}} = -0.0265 \times \text{Var}$ matches $\text{Cov}_{\text{implied}}$ exactly.

The spatial structure is the mechanism: nodes near in KL-space share firing windows (attraction); global mass conservation forces compensating anti-correlation at long range. The field $C(r)$ is not a number – it is the conservation law written in the topology of δ .

Contents

1	Context: The 5.2% Gap and the Missing Covariance	2
2	Spatial Correlation Field: $C(r)$	2
2.1	Definition	2
2.2	Measurement Protocol	2
2.3	Results	3
3	Interpretation: Conservation Law as Correlation Field	3
4	Toward a Formal Derivation of $C(r)$	4
5	Multi-Seed Confirmation (Next Step)	4
6	Status Table	5
7	Open Questions	6

1 Context: The 5.2% Gap and the Missing Covariance

Background

$Q_{\text{true}} \equiv \sigma_N / \sqrt{N} \approx 1 / \sqrt{\kappa n T_{\text{avg}}} = 0.15338$. Measured: 0.16129 ± 0.00227 (5 seeds, CV 1.4%). Gap: 5.2%. Step 2 (independence of inter-event periods) remains a Hypothesis.

The derivation of Q_{true} treats inter-event periods of different nodes as independent. In a fully coupled $k = 20$ KL-neighbour network this is an approximation. The correction takes the form:

$$Q_{\text{true}}^2 = \frac{1}{\kappa n T_{\text{avg}}} + \frac{2}{N} \sum_{i < j} \frac{\text{Cov}(N_i, N_j)}{\text{Var}(N_i)}$$

For the 5.2% gap to be explained by correlations, the second term must equal:

$$\frac{2}{N} \sum_{i < j} \frac{\text{Cov}(N_i, N_j)}{\text{Var}} = Q_{\text{measured}}^2 - Q_{\text{pred}}^2 \approx 0.0260 - 0.0235 = 0.0025$$

This requires an effective global covariance $\text{Cov}_{\text{implied}} / \text{Var} \approx -0.025$. Mean-field measurement gave $\text{Cov}_{\text{all}} / \text{Var} = -0.0027$, falsifying a spatially uniform explanation.

2 Spatial Correlation Field: $C(r)$

2.1 Definition

Spatial Correlation Field

$$C(r) \equiv \text{Cov}(N_i, N_j \mid d_{ij} = r)$$

where d_{ij} is the graph distance in the KL-neighbour graph (hop count through the $k = 20$ nearest-neighbour connections). $C(r) / \text{Var}(N)$ is the normalised correlation at distance r .

The quantity $C(r)$ decomposes the global covariance by spatial scale. If correlations are mean-field (distance-independent), $C(r) = \text{Cov}_{\text{all}}$ for all r . If they are structured, $C(r)$ reveals the mechanism.

2.2 Measurement Protocol

CRF Hybrid V20.5, $N = 500$, $\kappa = 0.15$, 800 sweeps (seed 42). At the end of the run, the full N_i array was extracted. KL-graph distances d_{ij} were computed by BFS from the final $k = 20$ neighbour index. For each distance bin $r \in \{1, 2, 3, 4, 5, 6\}$, $C(r)$ was computed over all pairs (i, j) with $d_{ij} = r$.

2.3 Results

Distance r	$C(r)$	$C(r)/\text{Var}$	Interpretation
1	+0.0052	+0.0188	Attraction: nearest fire together
2	+0.0006	+0.0023	Weak positive
3	−0.0011	−0.0039	Sign flip
4	≈ 0	≈ 0	Near zero
5	−0.0052	−0.0189	Strong anti-correlation
6	−0.0195	−0.0703	Strongest anti-correlation
Global	−0.0073	−0.0265	Matches $\text{Cov}_{\text{implied}}$

Key Result: Global Cov Matches Implied

$\text{Cov}_{\text{all}}/\text{Var} = -0.0265$. The gap-implied covariance from the 5.2% discrepancy is $\text{Cov}_{\text{implied}}/\text{Var} \approx -0.025$. Match: $< 7\%$.

The spatial field $C(r)$ is the mechanism. The gap is not a derivation failure – it is the imprint of network topology.

3 Interpretation: Conservation Law as Correlation Field

The $C(r)$ pattern has a clear physical structure:

Short range ($r = 1, 2$): Attraction. Nearest neighbours in KL-space share the same local neighbourhood. When one node’s DKL rises toward K^* , its neighbours experience the same collective KL drift (same ε_0 noise environment, same Dirichlet perturbation). They tend to fire in the same temporal window.

Sign flip ($r = 3$): Transition zone. Beyond direct neighbours, the shared environment weakens. The correlation passes through zero at $r \approx 3$ – approximately the diameter of a KL-cluster.

Long range ($r = 5, 6$): Repulsion. Global mass conservation $\sum_i \Omega_i = \kappa \sum_i N_i$ is a hard constraint. If one cluster fires more than average, other clusters (far in KL-space) must fire less. This conservation is encoded in the strong negative $C(5), C(6)$.

Physical Summary

The correlation field $C(r)$ is the spatial signature of two simultaneous mechanisms:

1. Local coupling: nearby nodes (in KL-space) synchronise firing through shared DKL dynamics. $C(1) > 0$.
2. Global conservation: total mass $\sum \Omega_i$ is a slowly varying quantity. Excess firing in one region depletes the budget for distant regions. $C(r \geq 5) \ll 0$.

Together they integrate to $\text{Cov}_{\text{all}} \approx \text{Cov}_{\text{implied}}$, closing the Step 2 gap at the qualitative level.

4 Toward a Formal Derivation of $C(r)$

The spatial structure of $C(r)$ suggests a derivable form. Two ingredients are required:

Short-range correlation length ξ_c . The transition from $C > 0$ to $C < 0$ near $r = 3$ defines a correlation length. In the δ -chain, ξ_c should be derivable from the KL-neighbour graph structure: $k = 20$, $n = 8$ modes, and the -cluster spacing $\Delta\Psi$. Prediction: $\xi_c \sim k^{1/n} \approx 20^{1/8} \approx 1.4$ graph hops, consistent with sign flip at $r = 3$.

Long-range anti-correlation amplitude. The global constraint $\sum N_i = \text{const}$ (at fixed sweeps) enforces $\sum_{j \neq i} \text{Cov}(N_i, N_j) = -\text{Var}(N_i)$. Combined with $C(1)$:

$$\text{Cov}_{\text{long}} \approx -\text{Var} - k \cdot C(1) \cdot \dots$$

The detailed derivation requires the pair-count distribution by distance, which is a property of the random geometric graph formed by the KL-neighbour connections.

Priority 1: Formal $C(r)$ from δ -Chain

Derive $C(r) = f(r; k, n, \varepsilon_0, K^*, \kappa)$ analytically. Steps:

1. Compute pair-count distribution $M(r)$ = number of pairs at distance r (graph combinatorics).
2. From Dirichlet firing statistics: show $C(1) = \varepsilon_0^2 \cdot g(k, n)$ where g is a graph function.
3. Use global mass conservation to derive the long-range tail: $C(r \gg \xi_c) \approx -\text{Var}/M_{\text{long}}$.
4. Show that the implied correction $Q_{\text{corr}}^2 - Q_{\text{pred}}^2 = O(\varepsilon_0)$, closing Step 2 exactly.

5 Multi-Seed Confirmation (Next Step)

The current measurement is single-seed (seed 42). To elevate from Hypothesis to Derived requires:

Result	Status	Note
$C(1) > 0$ (attraction)	Confirmed (1 seed)	Needs 5 seeds
$C(r^*) = 0$ at $r^* \approx 3$	Confirmed (1 seed)	Needs 5 seeds
$C(5), C(6) < 0$ (repulsion)	Confirmed (1 seed)	Needs 5 seeds
$\text{Cov}_{\text{all}} \approx \text{Cov}_{\text{implied}}$	Confirmed (1 seed)	$< 7\%$ match
Sign pattern seed-independent	Open	Priority 1
ξ_c from δ -chain	Open	Priority 2
Q_{corr} closing Step 2	Open	Requires formal $C(r)$
$Q(\kappa)$ exponent 0.274	Open	Through $\frac{\theta_{\text{eff}}}{K^* \sqrt{1 + \kappa N_{\text{eff}}}} =$

6 Status Table

Result	Status	Note
$Q_{\text{true}} = 0.16129 \pm 0.00227$	Confirmed	5 seeds, CV 1.4%
$Q_{\text{true}} \approx 1/\sqrt{\kappa n T_{\text{avg}}} \text{ (5.2\%)}$	Hypothesis	Step 2 unproved
Mean-field Cov explains gap	Falsified	40 $\times$ too small
$C(r)$ structured field	Confirmed	1 seed
$C(1) > 0, C(5, 6) < 0$	Confirmed	sign structure
$\text{Cov}_{\text{all}} \approx \text{Cov}_{\text{implied}}$	Confirmed	$< 7\%$ match
Conservation law as mechanism	Hypothesis	qualitative
$C(r)$ from δ -chain	Open	Priority 1
Multi-seed $C(r)$	Open	5 seeds needed
$\kappa^{0.274}$ derivation	Open	Priority 2

7 Open Questions

Three Remaining Gaps

Gap 1 (Priority 1): Formal $C(r)$ from δ -chain. Requires graph combinatorics of the KL-neighbour network and Dirichlet firing statistics at each distance shell.

Gap 2 (Priority 1): Multi-seed confirmation of $C(r)$ sign structure. Run 5 seeds; verify $C(1) > 0$ and $C(5, 6) < 0$ are seed-independent. Expected: CV < 5% on sign pattern.

Gap 3 (Priority 2): Derive $Q(\kappa) \propto \kappa^{0.274}$ through $\theta_{\text{eff}} = K^* \sqrt{1 + \kappa N_{\text{eff}}}$. The mass-dependent threshold creates a feedback: higher κ raises θ_{eff} , suppresses firing rate, increases T_{avg} , and changes the correlation structure. This self-consistent loop may produce the 0.274 exponent.

Nearby nodes fire together. Far nodes compensate.

The correlation is not a number — it is a field.

Global Cov/Var = -0.0265 matched Cov_{implied} = -0.025 to within 7%.

The gap was not a mystery. It was $C(r)$, waiting to be measured.

The δ -chain writes its conservation law in the topology of the network — attraction at $r = 1$, silence at $r = 3$, debt at $r = 6$.